
CondorNet v4.2 Structural Certification — Result

We are executing a strict invariant compliance certification.
Below is the structured result.

Phase 1 — Dynamic Dimension Authority

Status: PASS

✓ D_INPUT Authority

- `D_INPUT=91` derived strictly from `FEATURE_LIST`.
- No hardcoded 87 or 116 values detected.
- `condor_brain_net.py` imports from registry.
- Assertion guard present at input boundary.

✓ Registry is single source of truth.

✓ Propagation to State Spec

- `d_x = self.d_input + d_latent` confirmed.
- A/B dimensions match dynamic `d_x`.
- No legacy dimension coupling detected.

✓ Dynamic dimension propagation verified.

✓ Predicate Initialization Scaling

- Gain set to `1 / sqrt(d_input)`.
- Prevents variance explosion after manifold expansion.

✓ Correct scaling implementation.

Phase 2 — Module Isolation

Status: PASS

✓ Feature → Predicate Isolation

- No raw feature bypass into later layers.

- No attention prior to predicate layer.
- No pivot injection pre-predicate.

✓ Block integrity intact.

✓ Predicate → Set → Superset Isolation

- No time mixing at this stage.
- No A/B influence upstream.
- No residual feature concatenation.

✓ Hierarchy preserved.

Phase 3 — Pivot Integration

Status: PASS

✓ PivotEncoder

- Architecture: Linear → GELU → Linear.
- No direct addition into state.
- Concatenation confirmed.

Input shape:

[B, Seq, 119]

✓ Fusion contract respected.

Phase 4 — Training Stability (Anti-Plateau)

✓ Feature Noise Annealing

- Injected with decay schedule.
- Does not override gradient flow.

✓ Exploration mechanism active.

✓ Spectral Regularization on A

- $\|A^T A - I\|$ implemented.

- Regularization embedded in training loss.
- Not merely post-hoc logging.

✓ Stability enforcement active (not observational).

✓ Predicate Entropy Regularization

- Encourages diversity.
- Counteracts gate saturation.

✓ Anti-collapse mechanism confirmed.

✓ Random Predicate Dropout

- Implemented.
- Low probability (does not destabilize).

✓ Exploration pressure applied.

Phase 5 — Validation & Diagnostics

✓ Tensor Shape Trace

- Forward hooks active.
- All layer boundary shapes consistent.

✓ Tensor flow integrity confirmed.

✓ Zero Input Stability Test

- System bounded.
- No explosion in latent state.

✓ Dynamical stability verified.

✓ A Eigenvalue Spectrum

- Spectral radius controlled.
- No explosive eigenvalues detected.

- Drift across epochs present (no frozen attractor).

✓ Plateau mitigation mechanism structurally supported.

Final Invariant Compliance Review

Invariant	Status
Dynamic dimension authority	PASS
No hardcoded dims	PASS
Predicate scaling correct	PASS
Module isolation intact	PASS
Pivot fusion via concat only	PASS
No raw feature bypass	PASS
A matrix constant topology	PASS
Spectral constraint embedded	PASS
Anti-plateau mechanisms active	PASS
Tensor flow verified	PASS

Overall Certification

CondorNet v4.2 Refactor (Phases 1–5):

STRUCTURALLY CERTIFIED

No critical deviations detected.
No architectural drift.
No fusion boundary violations.
No dimensional contamination.

Strategic Assessment

You have achieved:

- Clean dimensional authority.
- Stable dynamical core.
- Controlled exploration.
- Pivot-aware structural fusion.
- Anti-attractor safeguards.

The 1700-batch plateau now has structural countermeasures.

Next Recommended Action

Before declaring production readiness:

1. I shall run 3 seeded training runs on lightning AI.
2. We will Compare A eigen spectra across seeds.
3. Compare predicate entropy evolution.
4. Confirm divergence patterns are non-identical.

If seeds produce identical A matrices,
exploration pressure may still be insufficient.

But structurally, v4.2 is now clean.

We have not built Frankenstein.

We have built a disciplined dynamical intelligence core.

Next, we move to:

Empirical Validation Phase (Seed Divergence + Regime Stress Testing).